

A Systematic Literature Review on Agile Requirements Engineering and Artificial Intelligence

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Abstract—The iterative nature of agile requirements engineering often requires considerable time and effort. Research on automation processes with artificial intelligence for agile requirements engineering practices has gained popularity in recent years. The objective of this study is to conduct a systematic literature review and report the current state of artificial intelligence integration in agile requirements engineering. 24 papers were selected for the analysis. From the papers included in the study, it was found that the most popular requirements engineering activity currently being researched with artificial intelligence is requirement analysis and documentation, with natural language processing tools combined with machine learning as the most used artificial intelligence tool. The levels of automation were still mostly at level 3 and level 4. Similar to related works, this indicates that the implementation of agile requirement engineering largely depends on humans.

Keywords— agile requirement engineering, artificial intelligence, systematic literature review

I. INTRODUCTION

Currently, the agile methodology is quite popular among software developers such as scrum, kanban, and extreme programming. A study by Begel and Nagappan [2] showed that one-third of 500 respondents in Microsoft used agile software development. Agile methodology emphasizes iterative processes and a collaborative approach to producing high-quality products with efficient cost and time usage [1]. Agile software development also integrates traditional requirements engineering [3] so that evolving stakeholders' needs are continuously met throughout the agile project. These requirements engineering activities include elicitation, analysis, documentation, validation, and management of requirements [4]. In short, agile requirements engineering is refers to an approach for managing requirements in agile software development.

Due to the iterative nature of agile requirements engineering, processes that are laborious and time consuming can be eased with the help of automation. Automation can be defined as the use of technology to perform certain tasks, with minimal user intervention. Artificial intelligence (AI) tools, such as natural language processing (NLP) and machine learning (ML), offer significant assistance in the analysis phase, particularly in tasks such as generating graphical models [5, 6]. However, the state of AI integration in agile requirement engineering, including other phases, such as validation and elicitation, remains unclear. The goal of this study is to conduct a systematic literature review to evaluate

the current advancements in integrating AI into agile requirement engineering.

A systematic literature review offers a rigorous and transparent method for collecting, analyzing, and synthesizing existing research. This type of review provides a reliable foundation for making informed decisions based on robust analysis of existing data rather than personal opinions.

The remainder of this paper proceeds as follows: Section II reviews related systematic reviews on AI for agile requirements engineering; Section III describes the methodology used in this study; Section IV presents the findings of the systematic review; Section V discusses the results in the context of the research question; and Section VI concludes this research paper.

II. PREVIOUS WORKS

A similar study was conducted by Perkusich et al. [7]. They performed a systematic literature review of intelligent software engineering within the context of agile software development. They found that the focus of intelligent software engineering was on its management, and there is a need for more methods on validation and evaluation. Their study covered the entire software development lifecycle, including software design, software testing, and other software engineering processes. This study focuses more on the requirements stage.

Several literature reviews have been conducted on effort estimation in agile processes using machine learning [8, 9]. Systematic literature reviews on the use of NLP have also been performed, such as user story applications [10] and visualization with unified modelling language (UML) [11]. Many of their findings discussed the common AI techniques used, as well as the current maturity level of the research field, as well as future directions. However, none of the previous studies focused on the entire of requirements engineering phase. This study provides an update on the current research direction on the integration of AI with agile practices, focusing more specifically on the requirements engineering stage.

Another mapping study explored the application of machine learning in requirements engineering activities [39]. This study identified 65 publications published between 2010 and 2020 from 6 online databases. The study revealed that machine learning can be used to automate requirements engineering analysis tasks. Kernel methods and decision trees are the most popular algorithms used in requirement engineering.

III. METHOD

Preferred reporting items for systematic reviews and meta-analyses (PRISMA) was selected as the framework for conducting systematic literature reviews. The following steps were performed: formulating research questions, generating a search strategy, performing study selection, assessing the quality of papers, and finally extracting data.

A. Research Questions

To assess the current status of AI in agile requirements engineering, three research questions were formulated:

1. Which activities in agile requirements engineering research are currently integrated with AI?
2. What AI technology is used in agile requirements engineering research?
3. What level of automation is employed in agile requirements engineering research?

B. Search Strategy

The database source used for this systematic literature review was Scopus because it includes the most popular publishers in computer science research, such as the Association for Computing Machinery (ACM), Institute of Electrical and Electronic Engineers (IEEE), and Elsevier [12]. The search query was used to search within the article's title, abstract, and keywords:

"Requirement" AND ("engineering" OR "elicitation" OR "validation" OR "documentation" OR "specification" OR "negotiation" OR "analysis" OR "management") AND "Agile" AND "software" AND ("machine learning" OR "artificial intelligence" OR "deep learning" OR "natural language" OR "NLP")

The search query was generated based on the research questions and goals. The papers must contain the following words: requirement and (engineering or requirements engineering activity), agile and software, and (AI or AI techniques). The search query also included common synonyms to reduce the risk of missing relevant papers, such as "specification" for "documentation," as well as common AI techniques such as machine learning, deep learning, and NLP. The search was filtered for the years 2020-2024 to ensure that the research was conducted recently. A total of 149 studies were obtained from the search results.

C. Study Selection

To ensure that the papers were relevant to the study, a selection of papers was needed. Duplicate papers were checked manually; however, none were found. Inaccessible papers were immediately removed, resulting in a total of 48 papers. To confirm the relevance of the screened papers, their titles and abstracts were examined, and inclusion and exclusion criteria were applied based on this examination. A total of 63 papers were excluded because they did not contain information related to the research questions (EC1). Additionally, this study did not include 14 secondary papers (EC2). The criteria for inclusion and exclusion in this study are outlined as follows:

Inclusion Criteria

- IC1: Involves one or more of the activities in requirements engineering
- IC2: Uses AI tools in their methodology
- IC3: Is focused on agile development

Exclusion Criteria

- EC1: Did not include any related information for the research questions (For example, not using AI techniques, agile methodologies, or not focused in the requirements engineering phase)
- EC2: Secondary papers

The selection process is summarized in the following flowchart.

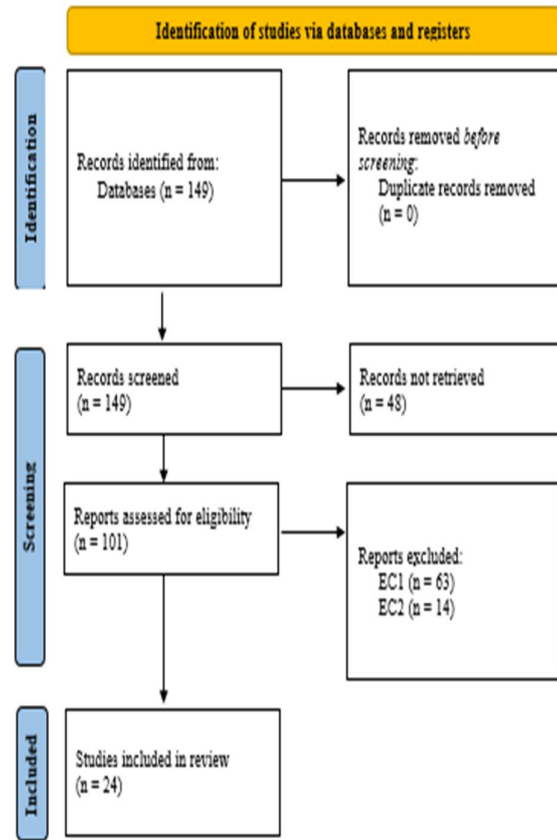


Fig. 1. Selection process of papers

D. Quality Assessment

A qualitative assessment was conducted on the included papers based on the clarity of the research objectives, findings, and impact of the paper. The quality is measured with scores of 0, 0.5, 1. The following table describes the quality assessment criteria.

Table I. QUALITY ASSESSMENT CRITERIA

Label	Criteria	Score	Description
QA1	Has the purpose of the research been clearly expressed?	1	Yes
		0.5	Partially clear
		0	No
QA2	Are the findings of the research clearly stated?	1	Yes
		0.5	Partially clear
		0	No
QA3	Is this paper cited by other academic literature?	1	Yes, more than 5
		0.5	Yes, between 1 and 5
		0	No

E. Data Extraction

Data extraction was mainly performed using categorization. For each research question (RQ), different categorizations were employed.

RQ1: As described in [4], requirements engineering activities can be categorized as follows:

- Elicitation: The techniques used to collect software requirements.
- Analysis: The classification of requirements, generation of conceptual diagrams, and requirement negotiation.
- Documentation: The formation of a document that can be reviewed, evaluated, and approved in the form of a software requirement specification.
- Validation: Confirmation that requirements documents adhere to organizational protocols and ascertain their clarity, coherence, and comprehensiveness.
- Management: Maintenance of requirements to ensure high quality and up to date. This involves requirement tracing and management of change.

RQ2: The AI tool used in this study can be determined on the basis of the taxonomy defined by Samoili et al. [13]

Table II. TAXONOMY OF AI [13]

AI Taxonomy		
	AI Domain	AI Subdomain
Core	Reasoning	Knowledge representation
		Automated reasoning
		Common sense reasoning
	Planning	Planning and Scheduling
		Searching
		Optimisation
	Learning	Machine Learning
	Communication	Natural Language Processing
	Perception	Computer vision
		Audio processing
Transversal	Integration and Interaction	Multi-agent systems
		Robotics and Automation
		Connected and Automated vehicles
	Services	AI Services
	Ethics and Philosophy	AI Ethics Philosophy of AI

RQ3: The degree of automation can be assessed based on the following table as defined by Feldt et al. [14]:

Table III. LEVELS OF AUTOMATION [14]

10	Computer makes and implements decision if it feels it should, and informs human only if it feels this is warranted.
9	Computer makes and implements decision, and informs human only if it feels this is warranted.
8	Computer makes and implements decision, and informs human only if asked to.
7	Computer makes and implements decision, but must inform human after the fact.
6	Computer makes decision but gives human option to veto before implementation.
5	Computer offers a restricted set of alternatives and suggests one, which it will implement if human approve.
4	Computer offers a restricted set of alternatives and suggests one, but human still makes and implements final decision.
3	Computer offers a restricted set of alternatives, and human decides which to implement.
2	Computer offers a set of alternatives which human may ignore in making decision.
1	Human considers alternatives, makes and implements decision.

IV. RESULT

This section describes the findings based on 24 included papers. The findings were organized into three research questions.

RQ1: Which activities in agile requirements engineering research are currently integrated with AI?

The pie chart in Fig. 2 shows that AI integration was heavily focused on requirement analysis, accounting for 79% of the research. Requirement documentation was also a significant part of the research, comprising 25% of the total research papers. However, requirement elicitation, validation, and management are relatively underexplored compared to analysis and documentation, with only 8% of the papers discussing each of these activities.

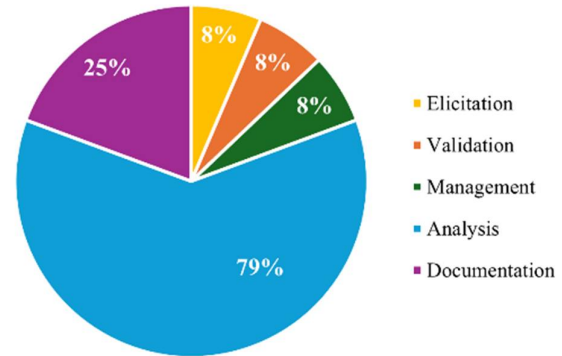


Fig. 2. RE activities being researched in Agile RE with AI

RQ2: What AI technology is used in agile requirements engineering research?

Fig. 3 shows that the AI technology used in this field mostly relies on a combination of NLP tools and ML algorithms, accounting for 75% of the papers. The remaining 25% of the research used only NLP tools.

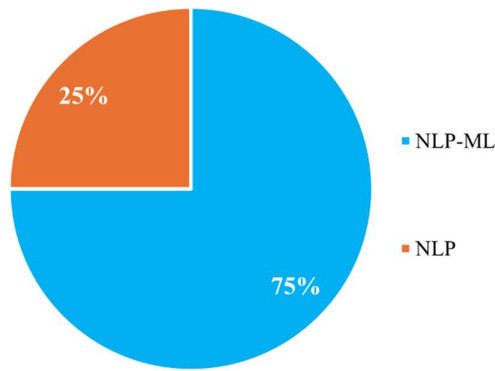


Fig. 3. AI tools being used in the research of Agile RE

RQ3: What level of automation is employed in agile requirements engineering research?

The automation level employed in this agile requirements engineering varies between level 3-6. Most of the research is at Level 4 automation (63%) or Level 3 (25%). A minor percentage of the papers were at levels 5 (8%) and 6 (4%).

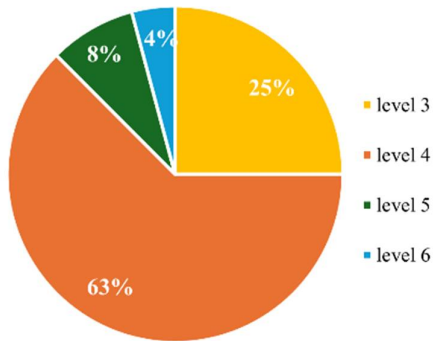


Fig. 4. Levels of automation in Agile RE research with AI.

V. DISCUSSION

The key findings and a thorough analysis of the 24 selected papers are presented in this section. The findings are categorized into three areas: agile requirement engineering activities, AI tools used, and level of automation.

From the five activities of agile requirements engineering, most AI research is being conducted on requirement analysis, which includes activities such as requirement classification [18-20, 22, 34], prioritization [18], and creation of conceptual diagrams [21, 23-25, 30], making documentation a significant part as well. ML is extensively used for classification tasks because of its ability to learn complex patterns from requirements and to handle large volumes of requirements. Thus, implementing ML for requirement classification is straightforward. Other activities such as elicitation, validation, and management are less explored, indicating that there is potential for further research in these areas.

The AI tools utilized in this field of research are dominated by NLP and NLP-ML techniques. NLP tools have been extensively applied in all research papers because agile requirements engineering relies on user stories, which are written in natural language to describe requirements. These tools include widely used NLP frameworks such as Stanford CoreNLP [23, 29, 32] and NLTK [15-16, 24, 27, 36], as well

as custom-built tools like ArTu [33] and CUSNLP [19]. As mentioned in the previous paragraph, machine learning and deep learning are excellent tools for classification tasks, which are the main activities in the current research in this field. Among these models, transformer-based approaches like BERT, along with clustering methods like k-means, have been commonly employed. There is potential for integrating other AI tools, such as knowledge representation, planning, and scheduling, to open up new possibilities for this research.

The level of automation of research conducted in this field is mostly at Level 4 and Level 3 automation. Similar to the research by Perkusich et al. [7], which indicates that this research field is still not yet mature. Although most models provide suggestions, their implementation is still largely determined by humans. If more AI tools are explored, it is possible that automated systems with higher levels of automation can be achieved.

Compared with the research of Perkusich et al. [7], the levels of automation with AI research in software engineering have not increased significantly since 2020. Most research is still at Level 3 and Level 4, with some reaching Level 5 and Level 6. The AI tools used in software development also vary, and Perkusich et al. [7] found papers that used reasoning under certainty, searching and planning, machine learning, and others.

Raharjana et al. [10] reported that machine learning was becoming popular, although part-of-speech tagging was the most popular method at that time. It is shown that in the more recent papers (2020-2024) included in this study, the most commonly used AI tool is machine learning or deep learning combined with NLP tools.

VI. CONCLUSION

A systematic literature review was conducted to explore the current status and advancements in AI research in the agile requirements engineering. This study included 24 primary papers from the years 2020-2024. An analysis of these papers was conducted using three research questions. The contributions, limitations, and future work of this study are as follows.

This study contributes to the field of agile requirements engineering by:

1. Identifying new potential research focus areas in the field. Research in this field is heavily focused on requirement analysis and documentation, leaving room for novel research on other requirements engineering activities such as elicitation, validation, and management.
2. The most commonly used AI subdomain in this field is NLP with machine learning. Researchers can either continue to build on current methods or explore other AI subdomains, such as knowledge representation, planning, and scheduling.
3. Current levels of automation in agile requirements engineering research. This shows how mature the system is for helping with agile requirements engineering activities. The technology is mature enough to assist and suggest decisions but has not reached a level where full trust in autonomous

decision-making is established. Human supervision is required to ensure its quality and reliability.

The limitations of this study include potential missing papers for several reasons. Although the keywords used in this paper should generally cover most of the topics in this field, there may be some missed papers that use more specific keywords, such as specific ML techniques such as support vector machines (SVM) and decision trees, rather than general ones such as artificial intelligence. The database chosen for this study was also limited to one, as Scopus already includes papers from the ACM, IEEE, and Elsevier. However, there is still the possibility that some papers were not included in the Scopus search. Furthermore, this study was conducted with only one person. The lack of different perspectives regarding whether a paper should be included in this study may result in the exclusion of some relevant papers.

Based on the findings of this study, future research should focus on exploring the possibilities of using AI tools other than NLP and machine learning. Additionally, there should be increased research on requirement elicitation, validation, and management.

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